

cURL & Checksum Verification Guide

URN Reference: [urn:nbn:de:101:1-2610041942043.302262298665](https://nbn-resolving.org/urn:nbn:de:101:1-2610041942043.302262298665)

Target Sources: <https://d-nb.info/1421225573/> & /34

Quick reference manual for securely downloading documents via `curl` and performing cryptographic integrity checksum verification in Termux.

1. Fetching Documents with cURL

To download files reliably while following redirects and handling headers from the German National Library portal or custom endpoints:

```
# Follow redirects (-L) and save output (-o) curl -L "https://d-nb.info/1421225573/34" -o document.pdf
```

2. Inspecting Download Headers

Before saving, use the `head` option to verify response codes (e.g., HTTP/2 200) and content types:

```
curl -I "https://d-nb.info/1421225573/34"
```

3. Cryptographic Checksum Validation

Ensure downloaded files are untampered and completely intact by generating local cryptographic hashes:

- **SHA-256 Checksum:**

```
sha256sum document.pdf
```

- **MD5 Checksum:**

```
md5sum document.pdf
```

4. Automated Integrity Guard Script

Integrate verification directly into shell scripts to abort if hashes mismatch:

```
EXPECTED_HASH="[Insert Expected SHA-256 Hash Here]" CURRENT_HASH=$(sha256sum document.pdf | awk '{print $1}') if [ "$CURRENT_HASH" = "$EXPECTED_HASH" ]; then echo "[+] Checksum verified successfully!" else echo "[!] ERROR: Checksum mismatch! File may be corrupted." exit 1 fi
```

